Meridian Wind Farm — Environmental Impact Assessment (Excerpt)

Project overview. The Meridian Wind Farm is located near Jaisalmer in Rajasthan. The project consists of 48 wind turbines with a total installed capacity of 144 megawatts, at 3 megawatts per turbine. The developer is Meridian Renewables Pvt Ltd. Construction began in January 2024 and commissioning is expected in December 2026. The total project cost is 1,150 crore rupees.

Turbine specifications. Each turbine is rated at 3 megawatts, with a hub height of 120 metres and a rotor diameter of 150 metres. The cut-in wind speed is 3 metres per second, the rated wind speed is 12 metres per second, and the cut-out wind speed is 25 metres per second. The wind farm is expected to generate 380 gigawatt-hours per year at a capacity factor of 30 percent.

Meridian Wind Farm — Environmental Impact Assessment (Excerpt) — Page 3

Bird and bat survey. A two-year survey carried out in 2022 and 2023 recorded 64 bird species in the project area. Collision risk modelling predicts 2.1 bird fatalities per turbine per year. Mitigation measures include curtailment during peak migration months, March to April and September to October, and radar-activated shutdown. Bat activity is highest in July and August.

Noise and shadow flicker. The predicted noise level at the nearest residence, 1.2 kilometres away, is 38 decibels (A-weighted), below the 45 decibel daytime limit and the 40 decibel night-time limit. Shadow flicker is predicted to reach a maximum of 18 hours per year at the nearest house, below the 30-hour guideline.

Meridian Wind Farm — Environmental Impact Assessment (Excerpt) — Page 5

Water and land use. The project covers 62 hectares in total, with a permanent footprint of 18 hectares. Water use during construction is 45,000 litres per day from a licensed borewell. As compensation, 12,000 native trees will be planted.

Community and jobs. Construction will create 320 jobs and operations will create 45 permanent jobs. A community benefit fund of 2.4 crore rupees per year will be established. 85 percent of the construction workforce will be hired locally. Grid connection is via a 220 kilovolt substation 8 kilometres away.